

MILESTONE-3

BACKEND:

A production-ready Flask + XGBoost web app predicting exoplanet habitability from 304 features across 33,799 planets from the NASA Exoplanet Archive.

Project Structure

```
'''
exo_habitability/
├── backend/
│   ├── app.py           ← Flask REST API (6 endpoints)
│   ├── utils.py         ← Preprocessing + feature alignment
│   └── models/
│       └── xgboost.pkl   ← Trained model
├── data/
│   ├── habitability_ranked.csv ← Full ranked dataset
│   ├── sample_planets.json  ← Curated UI samples
│   └── feature_cols.json    ← 304 model feature names
├── frontend/
│   ├── index.html        ← 3-page space UI
│   ├── style.css         ← Glassmorphism theme
│   └── script.js          ← Full API integration
└── requirements.txt
'''
```

Quick Start (PyCharm)

1. Install dependencies

```
'''bash
pip install -r requirements.txt
'''
```

2. Run the Flask server

```
'''bash
cd backend
python app.py
'''
```

3. Open in browser

```
'''
http://127.0.0.1:5000
'''
```

Flask serves both the API (`/api/*`) and the frontend (`/`) from one server.

API Endpoints

Method	Endpoint	Description
GET	<code>/api/health</code>	Health check + model status
POST	<code>/api/predict</code>	Single planet prediction
POST	<code>/api/predict/batch</code>	Batch prediction (max 500 rows)
GET	<code>/api/rank?n=10</code>	Top N most habitable planets
GET	<code>/api/sample</code>	Curated sample planets for UI
GET	<code>/api/sample/random</code>	One random planet from dataset
GET	<code>/api/features</code>	Model feature list (debug)

POST /api/predict — Example

```
```json
{
 "pl_rade": -0.673,
 "pl_bmasse": -0.187,
 "pl_orbper": 0.660,
 "pl_orbsmax": 0.875,
 "pl_eqt": -2.198,
 "pl_dens": -0.120,
 "st_teff": 0.201,
 "st_lum": 0.058,
 "st_met": -1.519,
 "st_spectype": "G0 V"
}
```

Response:

```
```json
{
  "habitable": 1,
  "probability": 97.42,
  "label": "Potentially Habitable",
  "confidence_band": "High",
  "status": "success"
}
```

POST /api/predict/batch — Example

Send a JSON array of up to 500 planet objects (same fields as above).

Response includes individual results + summary statistics.

Input Fields

- > ****Important****: All numeric values must be ****StandardScaler-normalised****
- > (as output by the training pipeline). Use the "Samples" or "Random" button
- > in the UI to auto-fill from the actual training dataset.

Field	Description	Example (scaled)
pl_rade	Planet radius	-0.673
pl_bmasse	Planet mass	-0.187
pl_orbper	Orbital period	0.660
pl_orbsmax	Semi-major axis	0.875
pl_eqt	Equilibrium temperature	-2.198
pl_dens	Planet density	-0.120
st_teff	Star effective temp	0.201
st_lum	Star luminosity	0.058
st_met	Star metallicity [Fe/H]	-1.519
st_spectype	Spectral type (string)	G0 V

How the 304-Feature Mapping Works

The model was trained on ****304 features****:

- 9 base planetary/stellar inputs (normalised)
- 4 engineered features (stellar compatibility, orbital stability)
- 291 One-Hot Encoded spectral type columns (drop_first=True)

`'utils.py'` handles this ****automatically****:

1. Takes your 9 numeric inputs + spectral type string
2. Computes derived features
3. Reads OHE column names ****directly from the model****
(`model.get_booster().feature_names`)
4. Sets `'st_spectype_G0 V = 1'`, all other spectral columns = 0
5. Aligns to exact 304-column model schema
6. Returns 1-row DataFrame ready for prediction

Batch CSV Format

The batch upload accepts CSV files with the same column names as the API:

```
```csv
pl_rade,pl_bmasse,pl_orbper,pl_orbsmax,pl_eqt,pl_dens,st_teff,st_lum,st_met,st_spectype
-0.673,-0.187,0.660,0.875,-2.198,-0.120,0.201,0.058,-1.519,G0 V
-0.444,-0.187,-0.212,-0.289,-1.207,-0.120,-1.555,0.058,-3.380,G0 V
```
```

Testing with curl

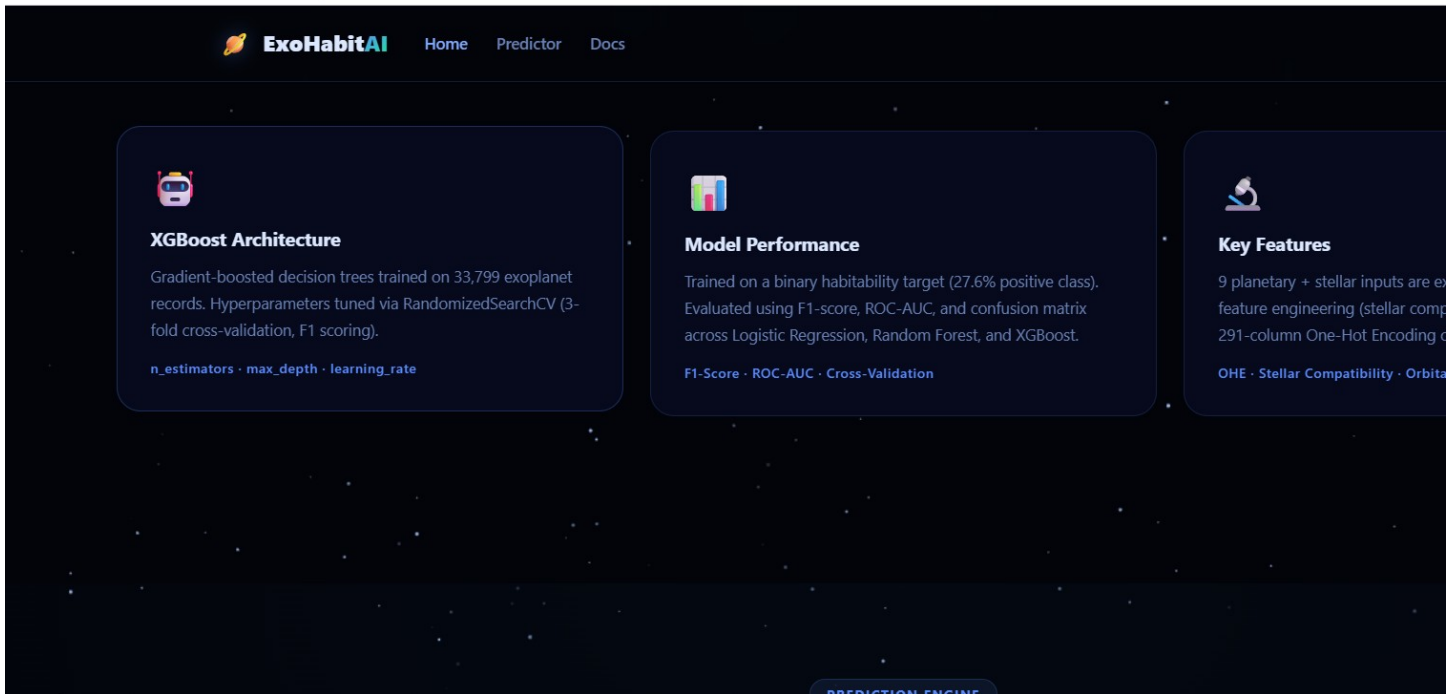
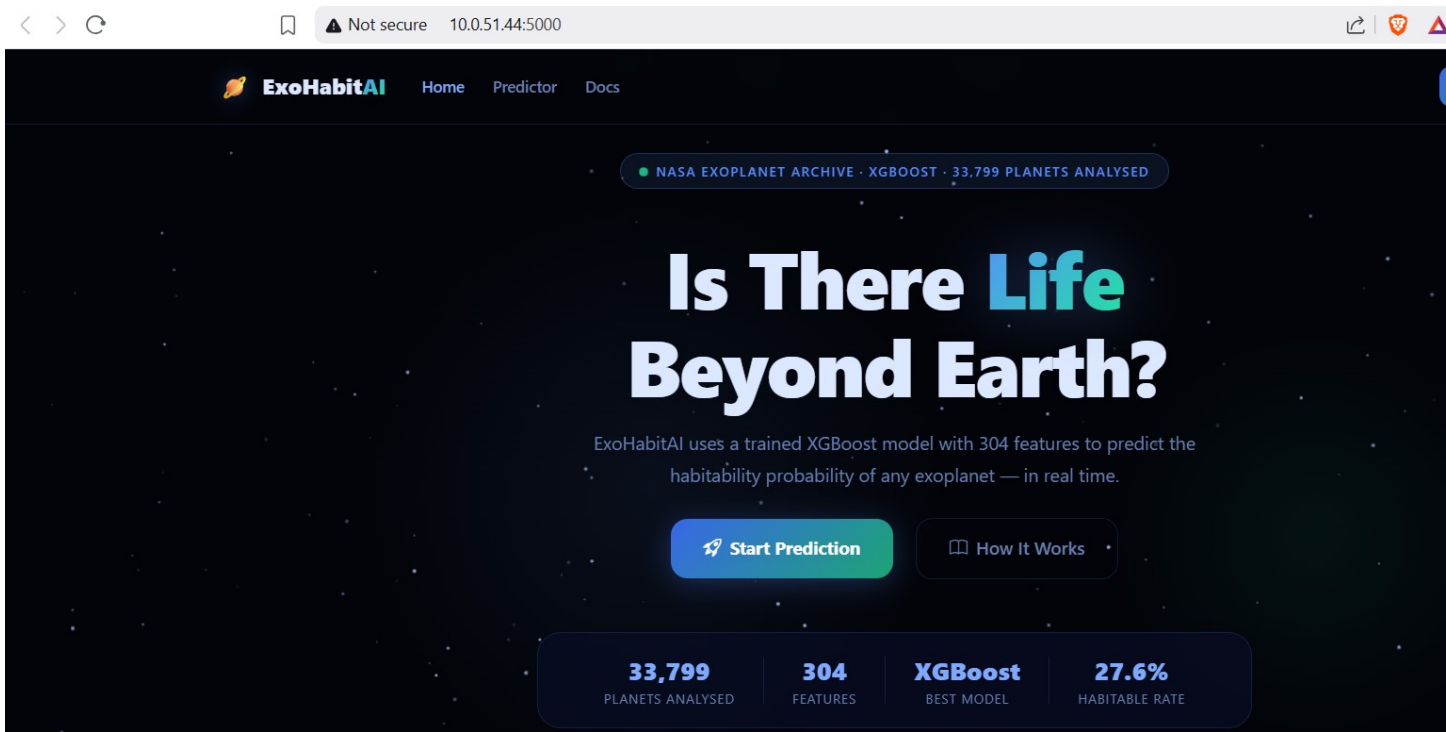
```
```bash
Health check
curl http://127.0.0.1:5000/api/health

Single prediction
curl -X POST http://127.0.0.1:5000/api/predict \
 -H "Content-Type: application/json" \
 -d '{"pl_rade":-0.673,"pl_bmasse":-0.187,"pl_orbper":0.66,"pl_orbsmax":0.875,"pl_eqt":-2.198,"pl_dens":-0.12,"st_teff":0.201,"st_lum":0.058,"st_met":-1.519,"st_spectype":"G0 V"}'

Top 10 planets
curl http://127.0.0.1:5000/api/rank?n=10

Random sample
curl http://127.0.0.1:5000/api/sample/random
```
```

FRONTEND:





Analyse Your Exoplanet

Enter planetary parameters or load a sample from the training dataset

Input Parameters

[Random](#)[Samples](#)

PLANETARY PROPERTIES

Planet Name (optional — any name you choose)

Radius (Earth radii, scaled)

Mass (Earth masses, scaled)

Orbital Period (days, scaled)

Semi-major Axis (AU, scaled)

Eq. Temperature (K, scaled)

Density (g/cm³, scaled)



1

Potentially Habitable

MODERATE CONFIDENCE

70.0%

HABITABILITY

e.g. G2 V, K5, M3 V, F8, G0 V

[New Scan](#)

BATCH UPLOAD



Drop a .csv or .json file here
Max 500 rows - Same fields as above

[Browse File](#)

Batch complete — sample_planets.json

Total Planets

12

Habitable

6

Not Habitable

6

Avg Probability

52.74%

[Download Results](#)

🔍 Top Habitable Exoplanets

| RANK | ID | PROB % | RADIUS | MASS | ORB.PERIOD | EQ.TEMP | STAR TEMP | SPECTYP |
|------|----------|--------|--------|--------|------------|---------|-----------|---------|
| 1 | EXO-0001 | 100.0% | -0.673 | -0.187 | 0.660 | -2.198 | 0.201 | G0 V |
| 2 | EXO-0002 | 100.0% | -0.662 | -0.187 | 0.660 | -2.198 | 0.199 | G0 V |
| 3 | EXO-0003 | 100.0% | -0.568 | -0.187 | 0.231 | -2.150 | 0.342 | G0 V |
| 4 | EXO-0004 | 100.0% | -0.574 | -0.187 | 0.388 | -2.198 | -0.730 | G0 V |
| 5 | EXO-0005 | 100.0% | -0.585 | -0.187 | 0.010 | -1.950 | 0.302 | G0 V |
| 6 | EXO-0006 | 100.0% | -0.579 | -0.187 | 0.024 | -1.713 | 0.702 | G0 V |
| 7 | EXO-0007 | 100.0% | -0.483 | -0.187 | 0.330 | -2.198 | 0.283 | G0 V |
| 8 | EXO-0008 | 100.0% | -0.486 | -0.187 | 0.330 | -2.198 | 0.283 | G0 V |
| 9 | EXO-0009 | 100.0% | -0.541 | -0.187 | 0.052 | -2.198 | -0.900 | G0 V |
| 10 | EXO-0010 | 100.0% | -0.623 | -0.187 | -0.159 | -1.753 | -1.051 | G0 V |

PROJECT DOCUMENTATION

Built With Science

End-to-end pipeline from raw NASA data to live AI predictions

Data Engineering

NASA Exoplanet Archive data loaded, 10 key features selected (planetary + stellar), duplicates and outliers removed, missing values imputed via median/mode, clipped at 1st–99th p

pandas

numpy

NASA Archive

Feature Engineering

Derived habitability index (temperature, radius, distance scores), stellar compatibility index, and orbital stability factor. Applied StandardScaler normalisation. One-Hot Encoded 291 s

OHE

StandardScaler

Feature Engineering

Tech Stack



Python 3



XGBoost



scikit-learn



pandas



NumPy



Flask



joblib



Bootstrap

API Reference

| | | |
|-------------|---------------------------------|---|
| POST | <code>/api/predict</code> | Single planet habitability prediction |
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| GET | <code>/api/sample</code> | Curated sample planets for auto-fill |